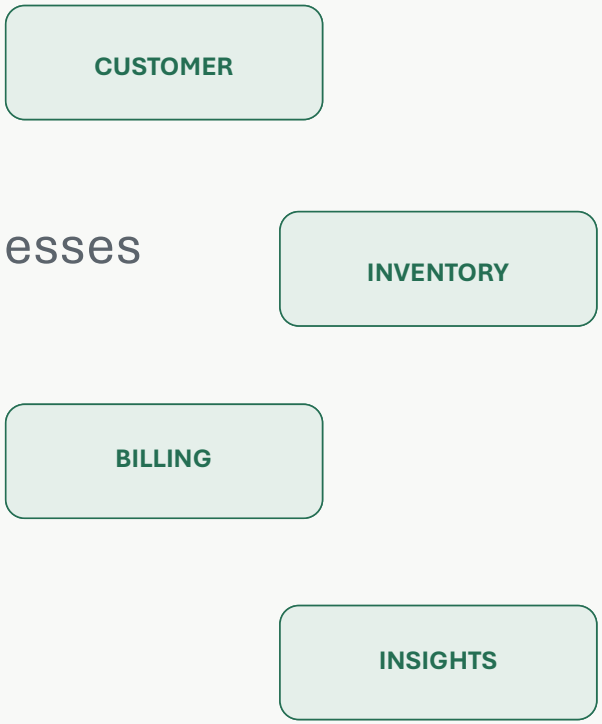
AI-NATIVE SOFTWARE PROJECT

BizPilot

An AI business assistant for India's small businesses

Sell faster. Control stock. Invoice safely.
Understand the business.



01 | THE PROBLEM

Small businesses run on memory, notebooks and missed messages.

The operational gap is not a lack of effort — it is a lack of an affordable digital assistant.

Slow replies

Customers ask on WhatsApp while the owner is serving someone else.

Stock uncertainty

Items run out unnoticed; slow-moving inventory ties up cash.

Manual billing

GST invoices and tax decisions are often handled manually.

No visibility

Owners know sales happened, but not always what is actually profitable.

02 | WHO WE SERVE

Built for the owner who does everything.

Kirana stores, medical shops, tailors, wholesalers and small online sellers.

Owner-operator

One person handles customers, purchasing, billing and decisions.

Low IT capacity

Tools must work without an IT team or lengthy training.

Local language

Hindi / Hinglish interaction can remove a major usability barrier.

Cost sensitive

Automation must be affordable at small-business margins.

Billing software records the business. BizPilot helps run it.

The differentiator is proactive, conversational assistance — with controls around money and customer data.

TODAY

Customer messages wait.
Stock is checked manually.
Invoices are prepared separately.
Business performance is guessed.

WITH BIZPILOT

AI responds.
Inventory is monitored.
Billing is generated from verified data.
Insights arrive automatically.

RESULT

Faster sales response
Fewer stock surprises
Safer operations
Better decisions

One assistant. Five specialized capabilities.

Customer Agent

Availability, pricing, FAQs and order capture.

Inventory Agent

Low-stock alerts, reorder suggestions and dead-stock signals.

Billing Agent

Invoice generation and verified tax-rule application.

Insights Agent

Sales, top products, trends and daily summaries.

Owner Control

Approvals for high-value orders, refunds and sensitive actions.

05 | HOW IT WORKS

From a customer message to a completed business action.

The AI reasons about intent, but deterministic tools remain the source of truth.

1

MESSAGE

“Do you have Tata Salt?”

2

ROUTE

Intent → Customer Agent

3

CHECK

Inventory + price tool

4

DECIDE

Rules + approval threshold

5

ACT

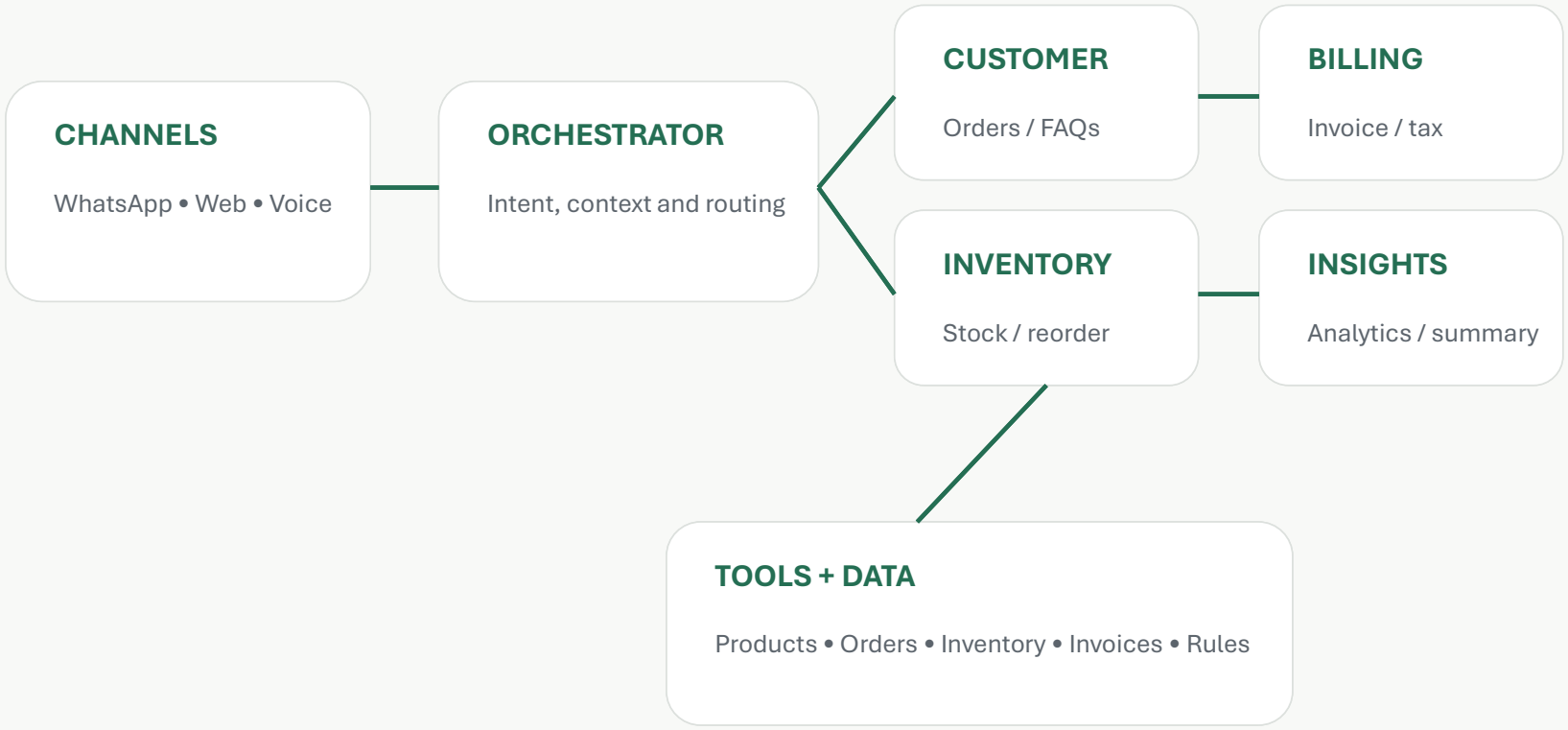
Create order / invoice

6

UPDATE

Inventory + owner summary

A controlled agent system — not a free-form chatbot.



07 | REALISTIC DAY

A day with BizPilot.

MORNING	**“3 items are low: sugar, Parle-G, Tata Salt.”**	Owner approves suggested supplier order.
AFTERNOON	**Customer asks: “Tata Salt available?”**	AI checks live stock, price and order rules.
EVENING	**Order crosses approval threshold.**	BizPilot pauses and requests owner approval.
CLOSE	**“Today's sales: ₹18,400.”**	Owner receives sales, best-seller and stock-risk summary.

AI can act — but it cannot act without boundaries.

The highest-risk actions are deliberately constrained.

GUARDRAILS

Block impossible discounts, invalid quantities and actions outside policy.

SECURITY

Protect customer PII, credentials, cost prices and tenant data.

HUMAN APPROVAL

Large orders, refunds and sensitive actions require the owner.

EVALUATION

Test expected and adversarial behaviour before release.

Modern enough for scale; simple enough for a student team.

FRONTEND

React + Tailwind CSS

Owner dashboard

BACKEND

Node.js + Express

REST APIs + auth

DATA

MySQL or MongoDB

Products, stock, orders

AI LAYER

LLM + tool calling

Agent orchestration

INTEGRATIONS

WhatsApp API / messaging

Email / supplier workflows

INFRASTRUCTURE

AWS + Nginx + PM2

Logs and monitoring

The project demonstrates the engineering around AI — not just the model.

Prompt & context engineering

Give agents only the context they need.

Agents & orchestration

Specialized agents use controlled tools.

RAG

Retrieve business policies / verified documentation when required.

Structured outputs

Force machine-readable decisions and actions.

Evals

Regression tests for accuracy, safety and tool selection.

Token optimization

Reduce context size and unnecessary model calls.

Observability

Trace prompts, tools, latency, errors and outcomes.

Security

Authorization, tenant isolation and prompt-injection defenses.

Prove value with measurable outcomes.

A pilot with one real shop can turn a concept into evidence.

RESPONSE TIME

Hours → seconds

Measure median customer response time.

OWNER TIME

Minutes saved / day

Track repetitive work automated.

STOCK

Fewer stock surprises

Compare stockout / dead-stock events.

QUALITY

Fewer invoice errors

Track billing corrections and exceptions.

Success metric: the system should be faster, safer and cheaper than the manual workflow it replaces.

Credibility comes from knowing where AI should stop.

Data quality

Stock and price data must stay accurate.

GST complexity

BizPilot supports verified rules; complex cases still need a CA.

Connectivity & APIs

WhatsApp access, approvals and connectivity are real deployment constraints.

Trust

Owners need gradual adoption, visibility and control before delegating actions.